Exercise 01:

1. How many different numbers can be represented with 16 bits?

   2^{16} = 65,536 numbers.

2. What is the largest 16-bit binary number that can be represented with

(a) Unsigned numbers?

(b) Two's complement numbers?

(c) sign/magnitude numbers?

   (a) $2^{16}-1 = 65535$; (b) $2^{15}-1 = 32767$; (c) $2^{15}-1 = 32767$

3. What is the smallest (most negative) 16-bit binary number that can be represented with

(a) Unsigned numbers?

(b) Two's complement numbers?

(c) sign/magnitude numbers?

   (a) 0; (b) $-2^{15} = -32768$; (c) $-(2^{15}-1) = -32767$

4. Convert the following unsigned binary numbers to decimal. Show your work.

(a) 1010_2

(b) 110110_2

(c) 11110000_2

(d) 000100010100111_2

   2215

   (a) 10; (b) 54; (c) 240; (d) ~~6311~~

5. Repeat the previous Exercise, but convert to hexadecimal.

   08A7

   (a) A; (b) 36; (c) F0; (d) 18A7

6. Convert the following hexadecimal numbers to decimal. Show your work.

(a) $A5_{16}$

(b) $3B_{16}$

(c) $FFFF_{16}$

(d) $D0000000_{16}$

(a) 165; (b) 59; (c) 65535; (d) 3489660928

7. Repeat the previous Exercise, but convert to unsigned binary.

(a) 10100101; (b) 00111011; (c) 1111111111111111;

(d) 11010000000000000000000000000000

8. Convert the following two's complement binary numbers to decimal.

(a) 1010_2

(b) 110110_2

(c) 01110000_2

(d) 10011111_2

(a) -6; (b) -10; (c) 112; (d) -97

9. Repeat the previous Exercise, assuming the binary numbers are in sign/magnitude form rather than two's complement representation.

(a) -2; (b) -22; (c) 112; (d) -31

10. Convert the following decimal numbers to unsigned binary numbers.

(a) 42_{10}

(b) 63_{10}

(c) 229_{10}

(d) 845_{10}

(a) 101010; (b) 111111; (c) 11100101; (d) 1101001101

11. Repeat the previous Exercise, but convert to hexadecimal.

(a) 2A; (b) 3F; (c) E5; (d) 34D

12. Convert the following decimal numbers to 8-bit two's complement numbers or indicate that the decimal number would overflow the range.

(a) 42_{10}

(b) -63_{10}

(c) 124_{10}

(d) -128_{10}

(e) 133_{10}

(a) 00101010; (b) 11000001; (c) 01111100; (d) 10000000; (e) overflow

13. Repeat the previous Exercise, but convert to 8-bit sign/magnitude numbers.

00101010; (b) 10111111; (c) 01111100; (d) overflow; (e) overflow

14. Convert the following 4-bit two's complement numbers to 8-bit two's complement numbers.

(a) 0101_2

(b) 1010_2

(a) 00000101; (b) 11111010

15. Repeat the previous Exercise if the numbers are unsigned rather than two's complement.

(a) 00000101; (b) 00001010

16. Base 8 is referred to as octal. Convert the following decimal numbers to octal.

(a) 42_{10}

(b) 63_{10}

(c) 229_{10}

(d) 845_{10}

(a) 52; (b) 77; (c) 345; (d) 1515

17. Convert each of the following octal numbers to binary, hexadecimal, and decimal.

(a) 42_8

(b) 63_8

(c) 255_8

(d) 3047_8

(a) 100010_2, 22_{16}, 34_{10}; (b) 110011_2, 33_{16}, 51_{10}; (c) 010101101_2, AD_{16}, 173_{10};
(d) 011000100111_2, 627_{16}, 1575_{10}

18. How many 5-bit two's complement numbers are greater than 0?

How many are less than 0? How would your answers differ for sign/magnitude numbers?

15 greater than 0, 16 less than 0; 15 greater and 15 less for sign/magnitude

19. How many bytes are in a 32-bit word? How many nibbles are in the word?

4, 8

20. A particular DSL modem operates at 768 Kbits/sec. How many bytes can it receive in 1 minute?

5,760,000

10^6

21. Hard disk manufacturers use the term "megabyte" to mean ~~106~~ bytes and "gigabyte" to mean ~~109~~ bytes. How many real GBs of music can you store on a 50 GB hard disk?

10^9

46.566 gigabytes

2^8

22. A memory on the Pentium II microprocessor is organized as a rectangular array of bits with ~~28~~ rows and ~~29~~ columns. Estimate how many bits it has without using a calculator.

128 Kbits $\rightarrow$ 2^9

23. Perform the following additions of unsigned binary numbers.
Indicate whether or not the sum overflows an 8-bit result.

(a) $10011001_2 + 01000100_2$

(b) $11010010_2 + 10110110_2$

(a) 11011101; (b) 110001000 (overflows)

24. Repeat the previous Exercise, assuming that the binary numbers are in two's complement form.

(a) 11011101; (b) 110001000

25. Convert the following decimal numbers to 6-bit two's complement binary numbers and add them. Indicate whether or not the sum overflows a 6-bit result.

(a) $7_{10} + 13_{10}$

(b) $17_{10} + 25_{10}$

(c) $-26_{10} + 8_{10}$

(d) $31_{10} + -14_{10}$

Exercise 02:

Here are some examples of **eight-bit**, twos complement binary addition.

In each case, we compute the sum, and note if there was an overflow.

-39 + 92 = 53

1	1		1	1				
	1	1	0	1	1	0	0	1
+	0	1	0	1	1	1	0	0
	0	0	1	1	0	1	0	1

No overflow and final C=1, so we neglect the carry. Sum is **positive**, equal to $2^5+2^4+2^2+2^0=53$.

104 + 45 = 149:

0	1	1		1				
	0	1	1	0	1	0	0	0
+	0	0	1	0	1	1	0	1
0	1	0	0	1	0	1	0	1

Overflow, and final C=0. Sum is not correct on 8 bits systems.

-75 + 59 = -16:

0	0	1	1	1	1	1	1	
	1	0	1	1	0	1	0	1
+	0	0	1	1	1	0	1	1
	1	1	1	1	0	0	0	0

No overflow and final C=0,
sum is 11110000 = -16

-19 + -7 = -26:

1	1	1	1	1			1	
	1	1	1	0	1	1	0	1
+	1	1	1	1	1	0	0	1
	1	1	1	0	0	1	1	0

No overflow and final C=1, so we neglect C.
Sum is **negative**; the value is 11111010= -26.

44 + 45 = 89:

		1		1	1			
	0	0	1	0	1	1	0	0
+	0	0	1	0	1	1	0	1
	0	1	0	1	1	0	0	1

No overflow nor carryout, sum is 89.

10 + -3 = 7:

1	1	1	1	1				
	0	0	0	0	1	0	1	0
+	1	1	1	1	1	1	0	1
	0	0	0	0	0	1	1	1

No overflow, final C=1, so we neglect the carry. Sum is correct and **positive=7**.

127 + 1 = 128:

0	1	1	1	1	1	1	1	
	0	1	1	1	1	1	1	1
+	0	0	0	0	0	0	0	1
	1	0	0	0	0	0	0	0

Overflow, C=0. Sum is not correct on 8 bits systems

-1 + 1 = 0:

1	1	1	1	1	1	1	1	
	1	1	1	1	1	1	1	1
+	0	0	0	0	0	0	0	1
	0	0	0	0	0	0	0	0

No overflow, final c =1, so we neglect the carry. Sum is correct and **positive**.

-103 + -69 = -172:

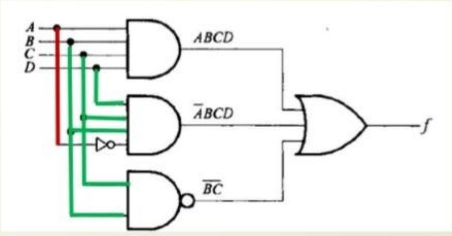
1		1	1	1		1	1	
	1	0	0	1	1	0	0	1
+	1	0	1	1	1	0	1	1
	0	1	0	1	0	1	0	0

Overflow, and C=1. Sum is not correct on 8 bits systems.

Exercise 03:

Example 1

- $f = ABCD + \bar{A}BCD + \overline{BC}$
- The logic diagram of this function is shown below



- $f = ABCD + \bar{A}BCD + \overline{BC}$
- The function can be simplified by using identities as follow:

$f = BCD(A + \bar{A}) + \overline{BC}$

$= BCD \,.\, 1 + \overline{BC}$ **using identity 4.a**

$= BCD + \overline{BC}$ **using identity 1b**

Assume BC = E $= ED + \bar{E}$

$= (E + \bar{E})(\bar{E} + D)$ **using identity 8b**

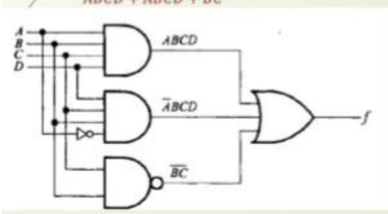
$= \bar{E} + D$ **using identity 4a**

Substitue E by BC $= \overline{BC} + D$

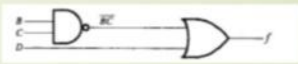
Example 1 (Cont'd)

- Therefore $f = ABCD + \bar{A}BCD + \overline{BC} = \overline{BC} + D$
- The logic diagrams of non-simplified and simplified function are equivalent

Logic diagram of non simplified function $ABCD + \bar{A}BCD + \overline{BC}$



Logic diagram of simplified function $\overline{BC} + D$



Example 2

- Show that $f = \overline{(a + \bar{b})(\bar{a} + b)}$ can be implemented using one Exclusive-OR (XOR)
  - Hint A XOR b = A ⊕ B = $\bar{A}B + A\bar{B}$

$f = \overline{(a + \bar{b})(\bar{a} + b)}$

$= \overline{(a + \bar{b})} + \overline{(\bar{a} + b)}$ **using identity 9.b (DeMorgan's Theorem)**

$= (\bar{a} \,.\, \bar{\bar{b}}) + (\bar{\bar{a}} \,.\, \bar{b})$ **using identity 9.a (DeMorgan's Theorem)**

$= \bar{a}b + a\bar{b}$ **using identity 5.a**

$= a \oplus b$

Example 3

- Show that $f = (X + \bar{X}Z)(X + Z)$ can be implemented using only one two-input OR gate.

$f = (X + \bar{X}Z)(X + Z)$	
$= (X + \bar{X})(X + Z)(X + Z)$	**using identity 8.b**
$= 1 \,.\, (X + Z)$	**using identities 4.a and 3.b**
$= X + Z$	**using identity 1b**

Example 4

- Express f for $\bar{f} = (\bar{A} + \bar{B} + \bar{C}) + \overline{ABC}$ using only one three input AND gate.

$f = \bar{\bar{f}} = \overline{(\bar{A} + \bar{B} + \bar{C}) + \overline{ABC}}$	**using identity 5.a**
$= \overline{(\bar{A} + \bar{B} + \bar{C})} \,.\, \overline{\overline{ABC}}$	**using identity 9.a**
$= (\bar{\bar{A}}.\bar{\bar{B}}.\bar{\bar{C}}).ABC$	**using identity 9.b**
$= ABC \,.\, ABC$	**using identity 3a**
$= ABC$	

Exercise 04:

1) Minimize each of the following expressions for F using a K-map:

Example 1 : $Y = \overline{A}\,\overline{B} + \overline{A}B + AB$

Solution:

A \ B	0	1
0	1 (0)	1 (1)
1	0 (2)	1 (3)

Simplified Expression : $Y = \overline{A} + B$

Example 2 : $Y = \overline{A}\,\overline{B}\,\overline{C} + \overline{A}B\,\overline{C} + A\overline{B}\,\overline{C} + A\overline{B}C + AB\overline{C} + ABC$

Solution:

A \ BC	00	01	11	10
0	1 (0)	0 (1)	0 (3)	1 (2)
1	1 (4)	1 (5)	1 (7)	1 (6)

Simplified Expression : $Y = A + \overline{C}$

Example 3: Y=

$\overline{ABCD} + \overline{AB}C\overline{D} + \overline{A}B\overline{C}D + \overline{A}BCD + AB\overline{C}D + ABCD + A\overline{BCD} + A\overline{B}C\overline{D}$

Solution:

AB \ CD	00	01	11	10
00	1 (0)	0 (1)	0 (3)	1 (2)
01	0 (4)	1 (5)	1 (7)	0 (6)
11	0 (12)	1 (13)	1 (15)	0 (14)
10	1 (8)	0 (9)	0 (11)	1 (10)

$$Simplified\ Expression: Y = BD + \overline{B}\,\overline{D}$$

Example 4: Y = F (A, B, C) = $\overline{A}B + ABC$

Solution

AB \ C	0	1
00		
01	1	1
11		1
10		

Simplified Expression: Y = $\overline{A}B + BC$

2) Simplify each of the following functions for F using a K-map:

Example 1: F (W, X, Y, Z) =Σ m (2, 3, 6,7, 8,9, 12, 13)

Solution

WX \ YZ	00	01	11	10
00			1	1
01			1	1
11	1	1		
10	1	1		

Simplified Expression: Y = $w\overline{y} + \overline{w}y$ *= w ▯ y*

Example 2 : F(W, X, Y, Z) = Σ m(1,3,5,7, 9, 11, 13, 15)

Solution

WX \ YZ	00	01	11	10
00		1	1	
01		1	1	
11		1	1	
10		1	1	

Simplified Expression: Y = Z

3) ***Minimize each of the following functions for f using a K-map and don't care conditions, d.***

Example 1 : F(A, B, C) = Σ m(1,2,4,7) And d(A, B, C) = Σ m (5,6)

Solution

AB \ C	0	1
00		1
01	1	
11	x	1
10	1	x

F= $A + \overline{B}C + B\overline{C}$

Example 2 : F(A, B, C, D) = Σ m(0,2,3,11) And d(A, B, C, D) = Σ m(1,8,9,10)

Solution

AB \ CD	00	01	11	10
00	1	x	1	1
01				
11				
10	x	x	1	x

F= $\overline{B}$

4) ***Find the essential prime implicants and then minimize each of the following functions for F using a K-map:***

Example 1: F(X, Y, Z) = Σ m (2, 3, 4, 5, 6)

Solution

XY \ Z	0	1
00		
01	1	1
11	1	
10	1	1

The essential prime implicants are the blue groups and the green group or the red one.

$\overline{X}Y + X\overline{Y} + Y\overline{Z}$ *OR* $\overline{X}Y + X\overline{Y} + X\overline{Z}$

Example 2: $wy + xy + \overline{w}xyz + \overline{wx}y + xz + \overline{xyz}$

Solution

$wy = \Sigma\, m(10,11,14,15)$

$xy = \Sigma\, m(6,7,14,15)$

$\overline{w}xyz = m7$

$\overline{wx}y = \Sigma\, m\,(2, 3)$

$xz = \Sigma\, m\,(5,7,13,15)$

$\overline{xyz} = \Sigma\, m\,(0, 8)$

yz \ wx	00	01	11	10
00	1			1
01		1	1	
11	1	1	1	1
10	1	1	1	1

The essential prime implicants are the colored groups: $F = y + xz + \overline{xz}$

5) ***Express F in the minimum product of sums form:***

$$F = \overline{R}S\overline{T} + R\overline{S}T + RST$$

Solution

RS \ T	0	1
00	0	0
01	1	0
11	0	1
10	0	1

$F = \prod M\,(0, 1, 3, 4, 6) = (R + S)(R + \overline{T})(\overline{R} + T)$

Exercise 05:

Example 2 : F(A, B, C, D) = Σm(0,2,3,11) And d(A, B, C, D) = Σm(1,8,9,10)

Solution

AB \ CD	00	01	11	10
00	1	x	1	1
01				
11				
10	x	x	1	x

F= $\overline{B}$

Exercise 2.31 Find a minimal Boolean equation for the function in Figure 2.86. Remember to take advantage of the don't care entries

A	B	C	D	Y
0	0	0	0	0
0	0	0	1	1
0	0	1	0	X
0	0	1	1	X
0	1	0	0	0
0	1	0	1	X
0	1	1	0	X
0	1	1	1	X
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	X
1	1	1	1	1

Figure 2.86 **Truth table for Exercise 2.31**

Solution:

$$Y = \bar{A}D + A\bar{B}\bar{C}\bar{D} + BD + CD = A\bar{B}\bar{C}\bar{D} + D(\bar{A} + B + C)$$

Exercise 2.28 Find a minimal Boolean equation for the function in Figure 2.85. Remember to take advantage of the don't care entries.

A	B	C	D	Y
0	0	0	0	X
0	0	0	1	X
0	0	1	0	X
0	0	1	1	0
0	1	0	0	0
0	1	0	1	X
0	1	1	0	0
0	1	1	1	X
1	0	0	0	1
1	0	0	1	0
1	0	1	0	X
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	X
1	1	1	1	1

Figure 2.85 Truth table for Exercise 2.28

Exercise 2.29 Sketch a circuit for the function from Exercise 2.28.

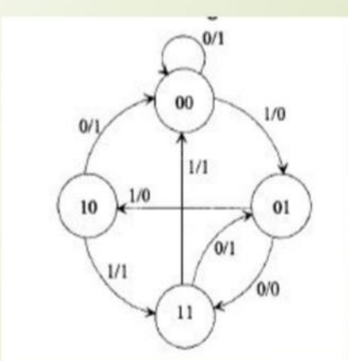
Solution:

Two possible options are shown below:

Exercise 06:

Example 5.2

Design a synchronous sequential circuit for the state diagram shown on the Figure, using D flip-flops.



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Example 5.2 Solution

- **Step 1:** Derive the state table from the state diagram.

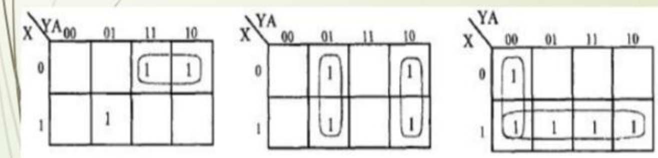
Present State		Input	Next State		Flip Flop Inputs		Output
X	Y	A	$X+$	$Y+$	D_X	D_Y	Z
0	0	0	0	0	0	0	1
0	0	1	0	1	0	1	0
0	1	0	1	1	1	1	0
0	1	1	1	0	1	0	0
1	0	0	0	0	0	0	1
1	0	1	1	1	1	1	1
1	1	0	0	1	0	1	1
1	1	1	0	0	0	0	1

In the state table, the next states are same as the flip-flop inputs because D flip-flops are used.

5

Example 5.2 Solution

- **Step 2 :** Derive the minimum forms of the equations for the flip-flop inputs. Using K-maps, the simplified equations for the flip-flop inputs can be obtained



(a) K-map for D_X $D_X = X\bar{Y}A + \bar{X}Y$

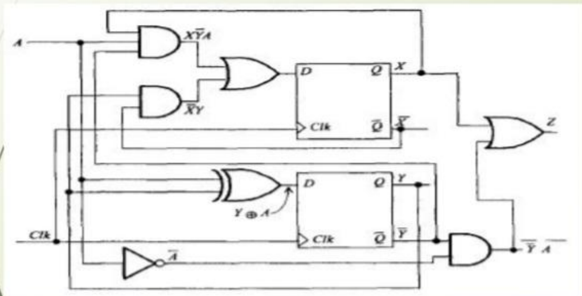
(b) K-map for D_Y $D_Y = \bar{Y}A + Y\bar{A} = Y \oplus A$

(c) K-map for Z $Z = \bar{Y}\bar{A} + X$

6

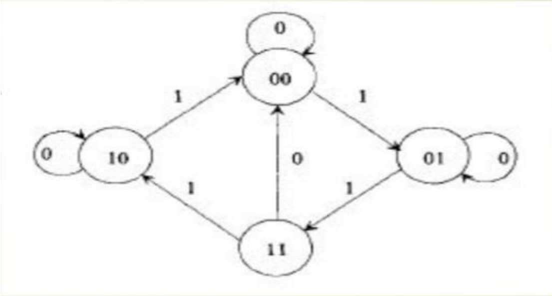
Example 5.2 Solution

- **Step 3:** Draw the logic diagram as shown in the figure.



Example 5.3

Design a synchronous sequential circuit for the state diagram of the figure using JK flip-flops.



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Example 5.3 Solution

- ***Step 1:*** Derive the state table from the state diagram.

Q	$Q+$	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Excitation Table of JK flip-flop

State Table

Present State		Input	Next State		Flip Flop Inputs			
X	Y	A	$X+$	$Y+$	J_X	K_X	J_Y	K_Y
0	0	0	0	0	0	X	0	X
0	0	1	0	1	0	X	1	X
0	1	0	0	1	0	X	X	0
0	1	1	1	1	1	X	X	0
1	0	0	1	0	X	0	0	X
1	0	1	0	0	X	1	0	X
1	1	0	0	0	X	1	X	1
1	1	1	1	0	X	0	X	1

9

Example 5.3 Solution

- The input ***A*** is 0 or 1 at each state, so the left three columns show all eight combinations for ***X*** , ***Y***, and ***A***.
- The next state column is obtained from the state diagram.
- The flip-flop inputs are then obtained using the excitation table for the JK flip-flop.
- For example, consider the top row. From the state diagram, the present state (00) remains in the same state (00) when input ***A*** = 0 and the clock pulse is applied.

Example 5.3 Solution

- ***Step 2:*** Derive the minimum forms of the equations for the flip-flop inputs. Using K-maps, the simplified equations for the flip-flop inputs can be obtained as shown:

X \ YA	00	01	11	10
0			1	
1	x	x	x	x

$J_X = YA$

X \ YA	00	01	11	10
0	x	x	x	x
1		1		1

$K_X = \overline{Y}A + Y\overline{A} = Y \oplus A$

(a) K-maps for J_X and K_X

X \ YA	00	01	11	10
0		1	x	x
1			x	x

$J_Y = \overline{X}A$

X \ YA	00	01	11	10
0	x	x		
1	x	x	1	1

$K_Y = X$

Example 5.3 Solution

- ***Step 3:*** Draw the logic diagram